

VIGNESH CHINTHAKUNTLA

AI Backend Developer | AI Automation Engineer | Prompt Engineer

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PROFESSIONAL SUMMARY

Results-oriented AI Backend Developer with experience building production-grade AI automation systems, LLM-powered agents, RAG pipelines, voice AI platforms, and workflow automation solutions. Skilled in Python, FastAPI, Django, n8n, Make.com, LangChain, AWS Bedrock, Vertex AI, Gemini, OpenAI, Pinecone, Microsoft Graph APIs, and cloud-based backend integrations. Strong expertise in prompt engineering, agent orchestration, AI workflow design, document intelligence, candidate evaluation automation, and enterprise productivity tools. Seeking AI-focused backend, automation, and prompt engineering roles involving scalable AI systems and business process transformation.

TECHNICAL SKILLS

- **Programming Languages:** Python, JavaScript, HTML, CSS, SQL
- **Backend Frameworks:** FastAPI, Django, REST APIs, Celery, Redis
- **AI & LLM Engineering:** OpenAI, Google Gemini, Vertex AI, AWS Bedrock, LangChain, RAG, Function Calling, Tool Calling, Agentic Workflows, Prompt Engineering
- **Vector DB & Search:** Pinecone, Typesense, RAG-based Search, Semantic Retrieval
- **Voice AI & Automation:** ElevenLabs, Twilio, AI Voice Agents, n8n, Make.com, Power Automate
- **Cloud & DevOps:** AWS S3, AWS Bedrock, GCP, Docker, Git, SharePoint, Microsoft Graph APIs
- **Databases:** PostgreSQL, MySQL, MongoDB, Redis, Google Sheets, Excel Automation
- **Business Tools & APIs:** Microsoft Teams, Outlook, SharePoint, SendGrid, Mailtrap, Zoho Mail Webhooks, SerpApi, Tableau, Power BI
- **Platforms:** Odoo, Shopify, Hostinger, One.com, WordPress

PROFESSIONAL EXPERIENCE

Sun Data Tech

Dec 2025 – Present

AI Backend Developer / Automation Engineer / Client: Prime Communications

- Serving as part of the IT engineering team for **Prime Communications**, supporting enterprise AI automation, hiring systems, Microsoft Teams tools, and AT&T-related training automation.
- **AI-Driven Applicant Tracking System:** Developed intelligent backend workflows for an AI-powered ATS covering candidate application intake, resume parsing, AI-based resume scoring, job recommendations, AI interview calls, video interview workflows, recruiter admin portal, candidate portal, offer letter generation, and automated candidate status handling.
- Built **CareerBot**, an AI job application assistant that allows candidates to apply faster using resume parsing and Easy Apply workflows. The system handles approximately **1,000 applications per day**, extracts job openings across **2,000+ Prime Communications stores**, and reduced candidate application time from minutes to seconds.
- Automated job distribution workflows across **Facebook Meta**, **LinkedIn**, and other recruitment channels using **Make.com** and backend automation pipelines, improving recruitment workflow speed by approximately **60%**.

- **Prime TeamsAssist App:** Built an internal AI assistant integrated within **Microsoft Teams** and **Outlook** to summarize emails, recommend actions, process meeting transcriptions, generate meeting summaries, and provide an Ask-AI interface for employees to get task-specific guidance.
- **Automated Invoice Processing System:** Designed and integrated an AI-powered invoice processing pipeline for multiple invoice formats and structures using **AWS, Microsoft Graph APIs, Outlook, Power Automate, SharePoint, AWS S3, and AWS Bedrock**. Built an end-to-end flow to extract invoice data, save structured outputs into Excel, and trigger respective Teams notifications.
- **AT&T FiberAssist Training Agent:** Built a multi-agent voice AI training platform on **ElevenLabs** to train sales representatives on AT&T Fiber. Designed an Orchestrator Agent for coaching, a Quiz Agent for skill assessment across six areas with silent scoring, and a Report Agent for performance breakdowns and personalized recommendations.
- Scaled the FiberAssist voice training system to **7,000+ monthly sessions**, reducing sales training delivery time by approximately **70%** and replacing traditional one-to-one coaching with an always-on conversational AI coach.

Webandcrafts

Dec 2024 – Dec 2025

Software Developer - R&D Team / Thrissur, Kerala

- Architected and developed AI-driven healthcare automation solutions using **Python, FastAPI, n8n, LangChain, Gemini, Vertex AI, Pinecone, and MongoDB**.
- Built an **Xpresshealth AI Automation Ecosystem** to automate healthcare staffing workflows including shift creation, bulk shift processing, staff communication, cancellations, swaps, onboarding, and support queries.
- Engineered intent-based AI agents to parse operational requests and trigger backend workflows, reducing manual processing latency from around **5 minutes to under 45 seconds**.
- Developed a RAG-based staff support system to answer payroll, pension, recruitment, sick leave, maternity leave, holiday, and policy-related queries using vector search and LLM-generated responses.
- Integrated **ElevenLabs, Twilio, and Google Live APIs** to automate voice-based staff coordination, availability checks, client approval flows, fallback queues, and shift confirmation workflows.
- Built asynchronous task pipelines using **Celery and Redis** for bulk data processing, improving backend responsiveness and system scalability.
- Migrated prototype workflows from **FastAPI** to **n8n** to accelerate experimentation, deployment, and business workflow iteration.

Indian Institute of Technology - Madras

Apr 2022 – Nov 2023

Full-Stack Developer Intern / Chennai, Tamil Nadu

- Contributed to the SGBC-IITM website by building responsive front-end interfaces using **HTML, CSS, Bootstrap, and JavaScript**, along with scalable backend systems using **Django**.
- Deployed the website using **AWS S3** and configured routing through **Route 53**.
- Managed and queried **MySQL** databases to support application workflows and reporting.
- Automated data-related tasks using **Excel macros**, pivot tables, and dashboards.
- Developed **Power BI** dashboards to visualize workflow processes and performed SEO optimization for the SGBC webpage.
- Project Link: [SGBC-IITM](#)

SELECTED PROJECTS

Legal Document AI Assistant (LegalRAG)

- Developed a full-stack **Retrieval-Augmented Generation (RAG)** application allowing users to query PDF and DOCX legal documents through both text and voice.
- Architected the backend using **FastAPI** and **SQLite**, with **JWT authentication** to ensure secure, per-user document namespaces.
- Implemented semantic search using **Pinecone** and integrated **Google Gemini 2.5 Flash** to generate accurate, context-grounded answers.
- Engineered interactive voice capabilities by integrating the **ElevenLabs Conversational AI widget** with dynamic session signing.
- Designed a modular architecture enabling users to search across their full document library or narrow queries to specific files.

QLoRA-Optimized SLM Resume Parser & Competency Mapper

- Developed a local-first recruitment automation pipeline using **QLoRA** to fine-tune compact Small Language Models such as **Qwen 2.5** and **SmolLM2**.
- Engineered the system to parse unstructured resumes into structured JSON and map extracted skills to ESCO-inspired competency categories using **sentence-transformer embeddings**.
- Generated quantitative job-fit scores by combining structured resume extraction, semantic skill matching, and competency mapping.
- Built the full ML lifecycle including automated data preparation, **AWS Bedrock** label enrichment, model evaluation, and inference optimized for consumer hardware.
- Deployed an interactive **Streamlit** UI to demonstrate real-time parsing, semantic skill matching, and robust JSON output repair.

AI Prompt Duplicate Detector

- Developed an NLP-based prototype in **Google Colab** to identify and filter semantically redundant Large Language Model prompts.
- Built a data processing pipeline using **Pandas** and **SentenceTransformers** to generate dense vector embeddings from textual inputs.
- Implemented cosine similarity algorithms using **scikit-learn** to detect and group near-duplicate prompts based on semantic meaning rather than exact keyword matches.
- Improved dataset curation and reduced downstream LLM processing costs by ensuring only high-quality, unique prompts are retained for training and inference.

Agentic Documentation Navigation System

- Developed an AI agent that can fetch, analyze, summarize, compare, and navigate large documentation systems based on user queries.
- Implemented RAG search using **Pinecone** and **Gemini Function Calling** to identify user intent and provide context-aware answers.
- Built version-aware document understanding to compare and respond based on older and newer versions such as V2.3 and V3.4.
- Designed for business use cases including law firms, news archives, technical documentation, medical reports, and policy documents.

n8n Workflow Automation Suite

- Built medical support chatbot workflows using **Gemini**, **Telegram Bot**, **MongoDB**, and memory-based personalization to support users based on medical history and chat context.
- Integrated **ElevenLabs** voice agents for mental support and escalation workflows.

- Developed Reddit automation workflows to retrieve trending posts, filter by engagement metrics, analyze content using AI, and identify business opportunities.
- Built niche-specific policy query chatbots using **Gemini**, **Pinecone**, **RAG**, memory, and email automation.

AI-Based Live Flood Detection and Alerting System for Tamil Nadu

- Implemented an AI-based flood risk monitoring system using Google Grounding, reservoir data, weather data, historical flood data, and satellite imagery.
- Designed alert workflows using **Twilio** to notify users automatically when flood risk crosses defined thresholds.

EDUCATION

Dr. M.G.R. Educational and Research Institute

2020 – 2024

Chennai, Tamil Nadu Bachelor of Technology in Computer Science Engineering - Artificial Intelligence

GPA: 8.5/10.0

CERTIFICATIONS

Advanced Computer Architecture	NPTEL, 2026
Full Stack Development Internship	Indian Institute of Technology - Madras, Nov 2023
Data Analytics Stack	Tableau, Power BI, SQL, Advanced Excel - Itvedant, 2024
Network Fundamentals	Cisco NetAcad, Jun 2023

LANGUAGES & PUBLICATIONS

- **Languages:** English, Telugu, Hindi, Tamil
- **Patent / Publication:** Developing efficient methods for tracking the location, status, and maintenance needs of oxygen cylinders [Springer] [IPI-202441000295]